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24k-0557, BSCS 1H.

ASSIGNMENT 3.

Q1). Ans:-

$y = x,\ y = 2x^2 - 1$

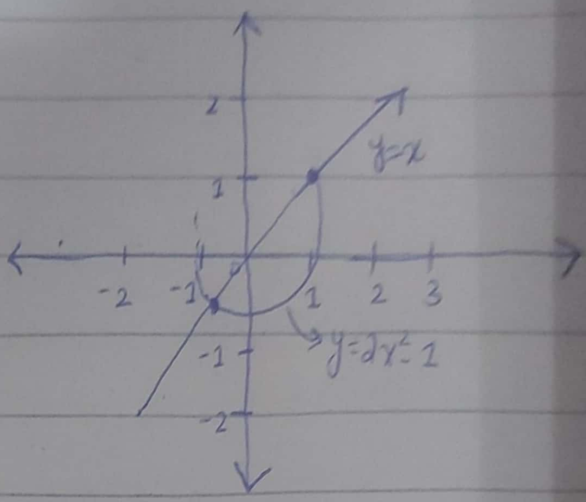
$x = 2x^2 - 1$

$2x^2 - x - 1 = 0$

$2x^2 - 2x + x - 1 = 0$

$2x(x-1) + 1(x-1) = 0$

$\boxed{x = 1,\ x = -1/2}$



$y = 2(1)^2 - 1 = 1$

$y = 2(-1/2)^2 - 1 = -1/2$

$V = \pi \int_{-1/2}^{1} f(x)^2 - g(x)^2 \rightarrow \int_{-1/2}^{1} (x)^2 - (2x^2-1)^2$

$(2x^2-1)(2x^2-1)$

$4x^4 - 4x^2 + 1$

$V = \pi \int_{-1/2}^{1} x^2 - [4x^4 - 4x^2 + 1]$

$V = \pi \int_{-1/2}^{1} -4x^4 + 5x^2 - 1 \rightarrow \boxed{\frac{9\pi}{20}}$

Integrate it $= \frac{9\pi}{20}$

Ans.

Q2). Ans:-

$\sum_{k=1}^{\infty} \frac{\sqrt{k}}{\sqrt{k^2+1}}$

$\frac{\sqrt{k}}{\sqrt{k^2(1+\frac{1}{k^2})}} \rightarrow \frac{\sqrt{k}}{k\left(\sqrt{1+\frac{1}{k^2}}\right)} = \frac{1}{\sqrt{k}\left(\sqrt{1+\frac{1}{k^2}}\right)}$ = it approaches 0

~~convergent~~

So it's inconclusive can be convergent or divergent

Q3). Ans:-

$\sum_{k=0}^{\infty} -ke^{-k}$

$\int -ke^{-k}$

$(-k)(-e^{-k}) - [-1 \times e^{-x}]$

$\int_{0}^{\infty} +ke^{-x} - e^{-x}$

$0 - 1 = \boxed{-1}$ → So it converges